Find the derivative of

$$f(x) = -6 \cdot \ln(\cos(7 \cdot x - 3))$$

Remember that you can ***always*** *leave notes in order to obtain points for your work even if your final answer is incorrect.*

Answer

$f'(x) =$ $-\frac{7\sin(7x-3)}{\cos(7x-3)}$ ✗ Wrong

Consider the two lines l and k given by

$$\begin{aligned} l &: \ 4 \cdot x - 2 \cdot p \cdot y = c \\ k &: \ -5 \cdot p \cdot x - 4 \cdot y = -3 \cdot c \end{aligned}$$

Where p and c are unknown numbers.

Find the value of p such that l and k intersect at the point $[-5, 2]$.

Determine the range of the following function:

$$f(x) = (4 \cdot \sin(8 \cdot x - 1) + 5)^2$$

Give your answer in one of the forms:

- $y > y_1 \wedge y < y_2$, or $y < y_1 \vee y > y_2$, or
- $y < y_1$ or $y > y_1$, or
- none if the range is empty, or
- all if the range consists of all numbers,

where y_1 and y_2 are exact numbers, and the symbols $<$ and $>$ can be replaced by $\leq$ and $\geq$ in the expressions above.

Answer

$y \geq 1 \wedge y \leq 81$ ✓ Correct

To get full points for this question, write your fully worked out solution in the note field.

Consider the function $f(x)$ given by

$$f(x) = -x^4 - 4 \cdot x^3 + 48 \cdot x^2 - 1$$

The function has inflection points, in one of these inflection points the function is increasing. Find the equation of the tangent line to $f(x)$ at this point.

Write your fully worked out solution in the note field (click the Note button to open the field)

Answer

$y =$ $112x - 81$ ✓ Correct

Consider the Lissajous curve C described by the functions

$$\begin{cases} x(t) &= -\frac{3 \cdot \sin(4 \cdot t + 1)}{9} \\ y(t) &= \frac{3 \cdot \cos(2 \cdot t - 6)}{4} \end{cases}$$

Find a time $\mathbf{t}$ such that $0 \leq t \leq \pi$ for which the speed of C is equal to $\frac{3}{2}$.

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Answer

$\mathbf{t} =$ - ✗ Wrong

Consider the function $f(x)$

$$f(x) = \frac{5}{x + 10} + \frac{5 \cdot x^2 + x + 8}{x + 4}$$

Find the oblique asymptote of $f(x)$ (and give its (simplified) function rule.

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Answer

$y =$ $5x - 19$ ✓ Correct

Consider the point $A = [1, 4]$ in the plane. Together with points B and C, it forms a triangle ABC such that the following three statements hold true:

1. $|AB| = 2 \cdot |BC|$
2. $|AC| = 3$
3. cos α ABC $= 3$

Give coordinates of points B and C.

Hint 1: Each of the properties gives a certain amount of points. If not all are satisfied, you can still get a partial grade. Try to start with an easy one.

Hint 2: There are infinitely many correct solutions.

Let $f(x)$ be given by

$$f(x) = (2 \cdot x - 3)^4 + \frac{2}{x} + \frac{3}{x^2}$$

Find an antiderivative, $F(x)$, of $f(x)$, such that $F(1) = 0$.

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Answer

$F(x) =$ $\frac{(2x-3)^5}{10} + 2\ln(x) + \frac{3}{x} + \frac{31}{10}$ ✗

You integrated the $(2 \cdot x - 3)^4$ term correctly, and the $\frac{2}{x}$ term partially correctly, the anti-derivative of $\frac{1}{x}$ is $\ln|x|$ and not $\ln(x)$. Moreover, the integration constant is not correct.

Consider the points $A = [6, 11]$ and $B = [-2, -5]$. Let $\mathbf{e}$ be the line connecting A and B, and let $\mathbf{f}$ be the line consisting of all the points $[x, t(x)]$ for which the following equation holds true:

$$d([x, t(x)], A) = d([x, t(x)], B)$$

This means that all the points on line $\mathbf{f}$ are equidistant from points A and B.

Give coordinates $[M_x, M_y]$ of a point M, such that there is a circle centered at M, with radius $R = 2 \cdot \sqrt{6}$, that is tangent to both $\mathbf{e}$ and $\mathbf{f}$.

Consider the function

$$f(x) = \frac{1}{\sqrt{4 - x^2}}$$

The area enclosed by the graph and the line $\mathbf{y} = 2$ is rotated around the $\mathbf{y}$-**axis** to form a solid of revolution B.

Calculate the volume of B and simplify as much as possible.

Remember that you can ***always*** *leave notes in order to obtain points for your work even if your final answer is incorrect.*

Answer

$V(B) =$ $\frac{9\pi}{2}$ ✓ Correct

Assume that $\mathbf{p}$ is a positive number, i.e. $p > 0$. Determine the domain of the family of functions

$$f_p(x) = \ln(x \cdot (20 \cdot p - 10) + 40 \cdot p - 5 \cdot x^2) - 1$$

Give your answer in one of the following forms:

- $x > x_1 \wedge x < x_2$, or $x < x_1 \vee x > x_2$, or
- $x < x_1$ or $x > x_1$, or
- none if the domain is empty, or
- all if the domain consists of all numbers,

where x_1 and x_2 are exact numbers, and the symbols $<$ and $>$ can be replaced by $\leq$ and $\geq$ in the expressions above.

Remember that you can ***always*** *leave notes in order to obtain points for your work even if your final answer is incorrect.*

Answer

Domain: $x \geq -2 \wedge x \leq 4p$ ✗ Your bounds are correct, but they should not be included in the domain.

On the charges of most of our electronic devices (phones, laptops, tablets, etc) there are transformers that convert alternate current (AC) of a given voltage to alternate current of a different voltage. On one such transformer, its output voltage as a function of time is described by $V(t)$ given by the following function rule

$$V(t) = 60 \cdot \sin\left(\frac{3 \cdot \pi}{5} \cdot t + \frac{7 \cdot \pi}{4}\right) \quad \text{Volts (V)}$$

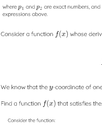
where $\mathbf{t}$ is measured in milliseconds (ms).

Determine its maximum voltage, average voltage, initial voltage (which is the voltage at $\mathbf{t} = 0$), and the time it takes between two consecutive maxima of the voltage in milliseconds. Simplify your answers as much as possible.

Answer

Maximum voltage:	$V_{max} =$	60	V	✓ Correct
Average voltage:	$\bar{V} =$	0	V	✓ Correct
Initial voltage:	$V_0 =$	$\sqrt{2} \cdot 30$	V	✗ Wrong

Consider a circle of radius 5 centered at the origin and imagine we want to construct a triangle AOB as seen in the picture below, with A on the x-axis and B on the circle, in such a way that $\angle OAB = \frac{\pi}{2}$ and $\angle BOA = \alpha$, with $0 < \alpha < \frac{\pi}{2}$.



Consider a line $\mathbf{a}$ given by:

$$x \cdot 4 + 2 \cdot y = 6$$

There is a line $\mathbf{t}$ that passes through the point $[0, 5]$, makes an angle of $\theta = \frac{\pi}{4}$ with $\mathbf{x}$, and has a positive slope.

Find the equation for $\mathbf{t}$. Give your answer in the form $\mathbf{y} = \mathbf{a} \cdot \mathbf{x} + \mathbf{b}$.

Let $f_p(x)$ be the following function:

$$f_p(x) = \begin{cases} -3 \cdot x^2 + 36 \cdot x - 81 & \text{for} \quad x < 0 \\ 5 \cdot x - 45 & \text{for} \quad 0 \leq x \leq 9 \\ 8 \cdot x + p x^2 - 8 & \text{for} \quad x \geq 9 \end{cases}$$

Give the value(s) of $\mathbf{p}$ for which $f_p(x)$ has a jumping discontinuity.

Give your answer in one of the forms:

- $p = p_1$ or $p = p_1 \vee p = p_2 \vee \cdots \vee p = p_n$,
- $p < p_1$ or $p > p_1$,
- $p < p_1 \wedge p > p_2$ or $p < p_1 \vee p > p_2$,
- none if there are no values of $\mathbf{p}$ for which $f_p(x)$ has a jumping discontinuity,
- all if $f_p(x)$ has a jumping discontinuity, for all values of $\mathbf{p}$,

where p_1 and p_2 are exact numbers and the symbols $<$ and $>$ can be replaced by $\leq$ and $\geq$ in the expressions above.

Consider the family of functions

$$f_p(x) = e^{x^2}$$

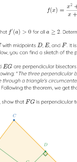
where $\mathbf{p}$ is a positive constant.

The tangent line $g_p(x)$ to $f_p(x)$ at the point $\frac{1}{p}$ is given by

$$g_p(x) = p \cdot e \cdot x$$

Calculate the area A which is enclosed by the functions $f_p(x)$, $g_p(x)$ and the $\mathbf{y}$-axis, as a function in $\mathbf{p}$.

Calculate the area A which is enclosed by the functions $f_p(x)$, $g_p(x)$ and the $\mathbf{y}$-axis, as a function in $\mathbf{p}$.



Consider the family of functions $f_p(x)$ given by

$$f_p(x) = (px)^2 + 6 \cdot p \cdot x - 2$$

assuming $p \neq 0$. Also consider the function $g(x)$ given by

$$g(x) = 2 \cdot x^2 - 8 \cdot x + 6$$

The function $g(x)$ has two roots. For which values of $\mathbf{p}$ is the x-coordinate of the vertex of $f_p(x)$ located between the two roots of $g(x)$?

Give your answer in one of the forms:

- $p > p_1 \wedge p < p_2$, or $p < p_1 \vee p > p_2$, or
- $p < p_1$, or $p > p_1$, or
- none if the domain is empty, or
- all if the domain consists of all numbers,

where p_1 and p_2 are exact numbers, and the symbols $<$ and $>$ can be replaced by $\leq$ and $\geq$ in the expressions above.

Consider a function $f(x)$ whose derivative is given by

$$f'(x) = x^3 - \frac{3 \cdot x^2}{2} - 6 \cdot x + 2$$

We know that the $\mathbf{y}$-coordinate of one of the inflection points of $f(x)$ is given by $\mathbf{y} = -6$.

Find a function $f(x)$ that satisfies these properties.

Consider the function

$$f(x) = \frac{x^2 + 5}{x + 3}$$

Assume $x \geq 2$ and use that $f'(x) > 0$ for all $x \geq 2$. Determine the inverse $f^{-1}(x)$

Consider the triangle ABC with midpoints D, E and F. It is given that DG is perpendicular to BC and EG is perpendicular to AB. Below, you can find a sketch of the problem.

The line segments DG and EG are perpendicular bisectors of sides BC and AB respectively. There is a theorem that states the following: *"The three perpendicular bisectors of a triangle intersect at the circumcenter. Thus, any line through a triangle's circumcenter and bisecting a side of the triangle must be perpendicular to that side."* Following the theorem, we get that FG should be perpendicular to AC.

Without using this theorem, show that FG is perpendicular to AC.